EXPERIMENT 10: IMPLEMENTATION OF SUBNETTING IN CISCO PACKET TRACER

AIM

To design multiple subnets, assign static IP addresses, connect the subnets through a router, and test packet transmission in Cisco Packet Tracer.

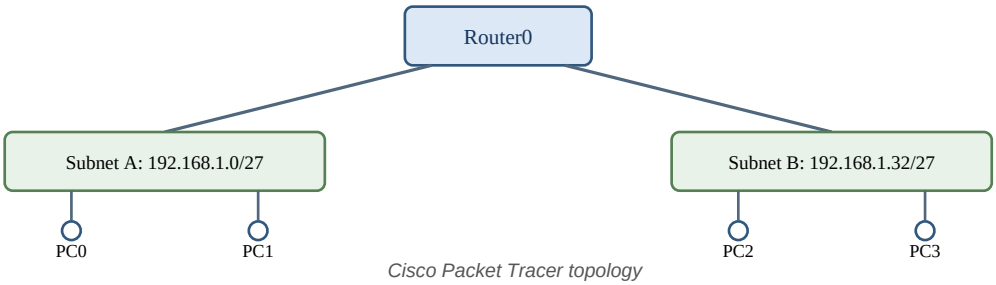
REQUIREMENTS

Cisco Packet Tracer, one router, two switches, four PCs, and copper straight-through cables.

THEORY

Subnetting divides a larger IP network into smaller logical networks. A /27 prefix uses subnet mask **255.255.255.224**, giving 32 addresses per subnet and 30 usable host addresses.

TOPOLOGY



Cisco Packet Tracer topology

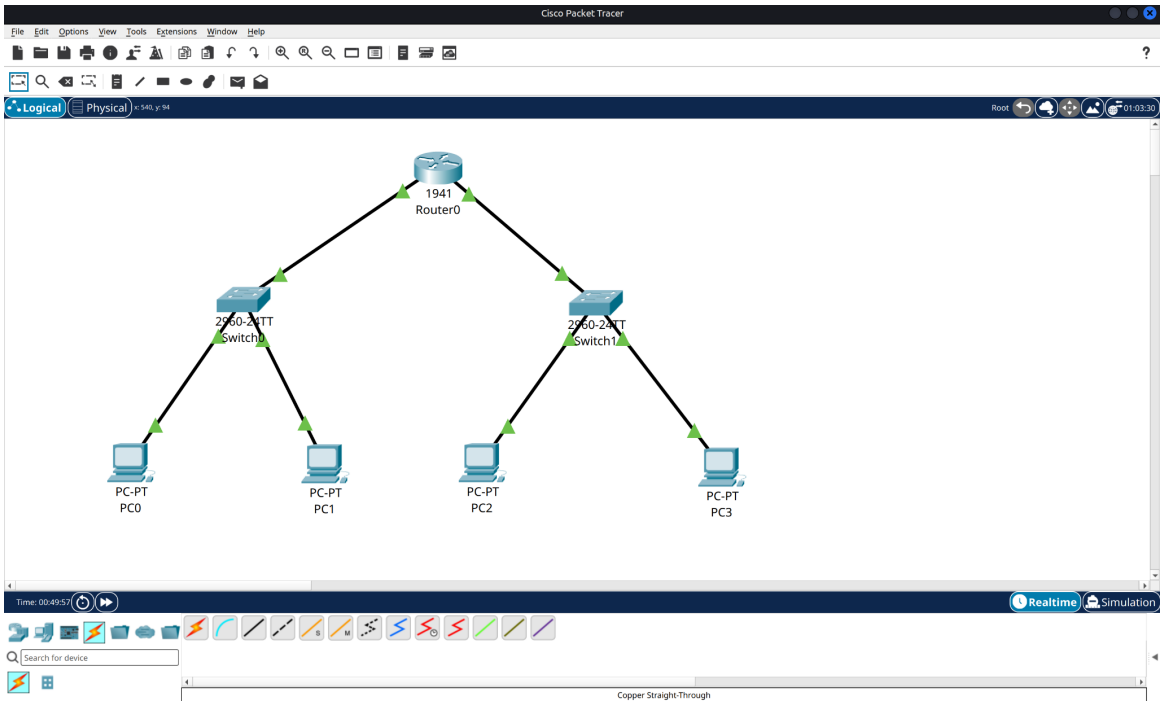


Figure: Configured two-subnet topology with active links.

PROCEDURE AND CONNECTIVITY TEST

1. Place one router, two switches, and four PCs in the Logical workspace. Connect each PC to its switch and connect each switch to a separate router interface using Copper Straight-Through cables.

2. Configure Router0 through the CLI:

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.224
no shutdown
exit
interface gigabitEthernet 0/1
ip address 192.168.1.33 255.255.255.224
no shutdown
end
```

3. Open each PC → Desktop → IP Configuration, select Static, and enter the corresponding IP address, subnet mask, and gateway.

IP ADDRESSING TABLE

Device	IPv4 address	Subnet mask	Default gateway
PC0	192.168.1.2	255.255.255.224	192.168.1.1
PC1	192.168.1.3	255.255.255.224	192.168.1.1
PC2	192.168.1.34	255.255.255.224	192.168.1.33
PC3	192.168.1.35	255.255.255.224	192.168.1.33
Router0 G0/0	192.168.1.1	255.255.255.224	—
Router0 G0/1	192.168.1.33	255.255.255.224	—

TESTING CONNECTIVITY

From PC0, open Desktop → Command Prompt and run the following command to test communication with PC2 in the other subnet:

```
ping 192.168.1.34
```

Observed output: 4 packets received, 0% packet loss. This confirms successful communication through Router0.

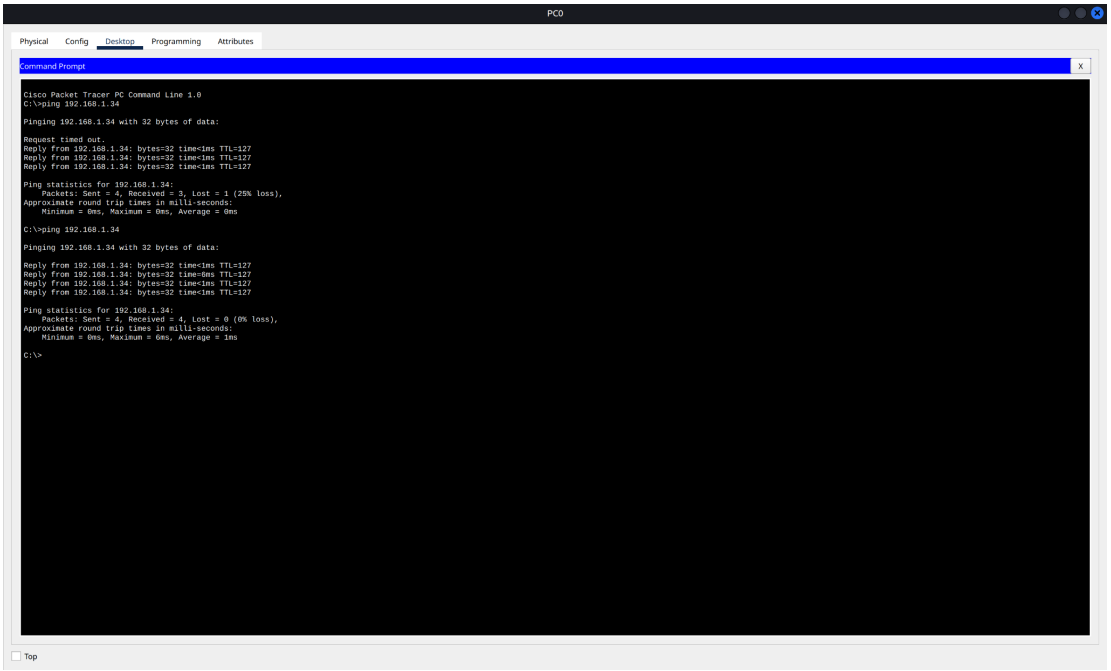


Figure: Successful ping from PC0 to PC2 (192.168.1.34).

OBSERVATIONS AND RESULT

a) What is subnetting?
Subnetting divides a large IP network into smaller logical networks using a subnet mask or prefix length.

b) What are the advantages of subnetting?
It uses IP addresses efficiently, reduces broadcast traffic, improves network organization, and supports easier administration and isolation.

c) Is subnetting implemented in your college?
Campus networks commonly separate departments, labs, Wi-Fi, and servers into different subnets. The actual subnet ranges depend on the college network configuration.

RESULT

Two /27 subnets were configured in Cisco Packet Tracer using Router0, two switches, and four PCs. Static IP addresses and default gateways were assigned, and successful packet transmission between the subnets was verified using ping.